

A basis for efficient representation of the S-transform

R.G. Stockwell

Northwest Research Associates, Colorado Research Associates Division, 3380 Mitchell Lane, Boulder, CO 80301, USA

Available online 19 May 2006

Abstract

The S-transform is a time-frequency representation known for its local spectral phase properties. A key feature of the S-transform is that it uniquely combines a frequency dependent resolution of the time-frequency space and absolutely referenced local phase information. This allows one to define the meaning of phase in a local spectrum setting, and results in many desirable characteristics. One drawback to the S-transform is the redundant representation of the time-frequency space and the consumption of computing resources this requires (a characteristic it shares with the continuous wavelet transform, the short time Fourier transform, and Cohen's class of generalized time-frequency distributions). The cost of this redundancy is amplified in multidimensional applications such as image analysis. A more efficient representation is introduced here as a orthogonal set of basis functions that localizes the spectrum and retains the advantageous phase properties of the S-transform. These basis functions are defined to have phase characteristics that are directly related to the phase of the Fourier transform spectrum, and are both compact in frequency and localized in time. Distinct from a wavelet approach, this approach allows one to directly collapse the orthogonal local spectral representation over time to the complex-valued Fourier transform spectrum. Because it maintains the phase properties of the S-transform, one can perform localized cross spectral analysis to measure phase shifts between each of multiple components of two time series as a function of both time and frequency. In addition, one can define a generalized instantaneous frequency (IF) applicable to broadband nonstationary signals. This is the first time a channel IF has been integrated in an orthogonal local spectral representation. A direct comparison between these basis functions and complex wavelets is performed, highlighting the advantages of this approach. The relationship between this basis set and the fully redundant S-transform is demonstrated highlighting the ability to arbitrarily sample the time-frequency space. The introduction of this basis set leads to efficient analysis routines that may find use in a wide range of fields.

© 2006 Elsevier Inc. All rights reserved.

Keywords: S-transform; Time-frequency analysis; Wavelets

1. Introduction

In observations of the ocean and atmosphere, one finds that the environment is full of waves that are obviously not infinite in extent nor constant over time. For instance, acoustic waves, internal waves, tides and tidal harmonics, inertial oscillations, surface and other interface waves, solitons and nonlinear wave events exhibit changing characteristics. Many of these waves, however, have an obvious oscillatory manner, which have lead over the years to many attempts to analyze the dynamic spectrum or local spectral nature of the observations.

One popular method is the short time Fourier transform [1] and the related Gabor transform [2]. A closely related method is complex demodulation [3] which produces a series of band pass filtered voices and is also related to the

E-mail address: stockwell@co-ra.com.

filter bank theory of wavelets. Another family of time-frequency representations is the Cohen class [4] of generalized time-frequency distributions (GTFD). Because of the quadratic nature of this class, the presence of cross terms can cause difficulty in the interpretation. Some particular choices for kernel can help to alleviate these problems. Specific GTFDs include the Wigner distribution, the Cone–Kernel distribution [5], the Choi–Williams distribution [6] as well as the smoothed pseudo Wigner distribution (PWD) [7] which reduce the effect of cross terms, but in doing so fail to satisfy the marginal properties of the Wigner distribution.

One of the more popular attempts to localize the spectrum has been with the continuous wavelet transform [8]. The basic philosophy is that any time series can be decomposed into a self-similar series of dilations and translations of a mother wavelet. Strictly, the continuous wavelet transform gives a representation in the state space of dilation a and translation b , but with the appropriate choice of wavelet, it can be used to measure the power spectrum locally.

In the effort to find an orthonormal and yet local set of basis functions, the discrete orthonormal wavelet transform (DWT) has been discovered by Daubechies [9]. Orthonormal functions are those which are both orthogonal and normalized (for basis functions u and v , $u \cdot v = 0$, and $u \cdot u = v \cdot v = 1$). The DWT provides a picture of a time series that gives very good time resolution, and some insight into the spectral components present. However, the information is sparse and the spectral regions are not unique (different dilations have overlapping spectral responses). The DWT using Daubechies wavelets is a powerful tool useful in many applications (e.g., image compression, etc.), however, a clearer time-frequency representation is desired. There have been many good review papers in the literature by Daubechies [9], Young [10], Rioul and Vetterli [11], Hlawatsch and Boudreaux–Bartels [7], Cohen [4], and Qian [12].

The role of phase in wavelet analysis is not as well understood as it is for the Fourier transform, especially for orthonormal wavelet representations. Current complex wavelets such as complex Daubechies wavelets [13], dual tree complex wavelets [14–16], and complex Shannon wavelets [17] do not have a direct relationship to the Fourier transform. Since “complex Daubechies wavelets are not analytic, there is not a direct equivalence between the Fourier and wavelets as to the meaning and the role of the phase,” Lina [18].

A more recent time-frequency representation, the S-transform [19], has found application in a range of fields. It is similar to a continuous wavelet transform in having progressive resolution but unlike the wavelet transform the S-transform retains absolutely referenced phase information. Absolutely referenced phase information means that the phase information given by the S-transform refers to the argument of the sinusoid at zero time (which is the same meaning of phase given by the Fourier transform). The S-transform defines what local phase means in an intuitive way, at a peak in local spectral amplitude (indicating a quasimonochromatic signal), as well as off peak, where the rate of change of the phase leads to a channel instantaneous frequency analysis. The S-transform not only estimates the local power spectrum, but also the local phase spectrum. It is also applicable to the general complex valued time series.

One drawback to the S-transform is the size of its redundant representation of the time-frequency plane. It is apparent that a more efficient representation of the S-transform is needed, one that provides a framework on which reduced sampling can be laid. This is the objective of this manuscript.

2. The S-transform

The continuous S-transform [19] of a function $h(t)$ is

$$S(\tau, f) = \int_{-\infty}^{\infty} h(t) \frac{|f|}{\sqrt{2\pi}} e^{-(\tau-t)^2 f^2/2} e^{-i2\pi f t} dt. \quad (1)$$

A voice $S(\tau, f_0)$ is defined as a one-dimensional function of time for a constant frequency f_0 , which shows how the amplitude and phase for this exact frequency changes over time.

In the discrete case, there are computational advantages to using the equivalent frequency domain definition of the S-transform (where $H[\frac{n}{NT}]$ is the Fourier transform of the N -point time series $h[kT]$)

$$S\left[jT, \frac{n}{NT}\right] = \sum_{m=0}^{N-1} H\left[\frac{m+n}{NT}\right] e^{-2\pi^2 m^2/n^2} e^{i2\pi m j/N}, \quad n \neq 0, \quad (2)$$

and for the $n = 0$ voice, it is equal to the constant defined as

$$S[jT, 0] = \frac{1}{N} \sum_{m=0}^{N-1} h[mT], \quad (3)$$

where j , m , and $n = 0, 1, \dots, N - 1$. The sampling of the S-transform is such that $S[jT, n/NT]$ has a point at each time sample and at each Fourier frequency sample. Similar to a STFT and CWT, this is redundant. The discrete inverse of the S-transform is found by averaging over time to get the Fourier transform spectrum, and inverting to the time domain

$$h[kT] = \sum_{n=0}^{N-1} \left\{ \frac{1}{N} \sum_{j=0}^{N-1} S \left[jT, \frac{n}{NT} \right] \right\} e^{i2\pi nk/N}. \quad (4)$$

The utility of S-transform analysis of a time series has been demonstrated in a range of geophysical fields such as: (i) analyzing internal atmospheric wave packets [20,21], (ii) atmospheric studies [22–26], (iii) characterization of seismic signals [27–30], (iv) global sea surface temperature analysis [31,32], as well as: (v) electrical engineering [33,34], (vi) mechanical engineering [35,36], and (vii) digital signal processing [37–40]. The S-transform has also been used in the medical field, in (viii) human brain mapping [41], (ix) cardiovascular studies [42], (x) MRI analysis [43], and (xi) the physiological effects of drugs [44].

2.1. Why use the S-transform approach?

What distinguishes the S-transform from the many time-frequency representations available is that the S-transform uniquely combines progressive resolution with absolutely referenced phase information. Daubechies has stated that progressive resolution gives a fundamentally more sound time-frequency representation [9]. Absolutely referenced phase means that the phase information given by the S-transform is always referenced to time $t = 0$, which is also true for the phase given by the Fourier transform. This is true for each S-transform sample of the time-frequency space. This is in contrast to a wavelet approach, where the phase of the wavelet transform is relative to the center (in time) of the analyzing wavelet. Thus as the wavelet translates, the reference point of the phase translates. This is called “locally referenced phase” to distinguish it from the phase properties of the S-transform.

From one point of view, local spectral analysis is a generalization of the global Fourier spectrum. In fact, since no record of observations is infinite, all discrete spectral analysis ever performed on measured data have been local (i.e., restricted to the time of observation). Thus there must be a direct relationship between a local spectral representation and the global Fourier spectrum. This philosophy can be stated as the fundamental principle of S-transform analysis:

The time average of the local spectral representation should result identically in the complex-valued global Fourier spectrum.

This leads to phase values of the local spectrum that are obvious and meaningful. While there can be several important definitions of “proper phase,” a natural one would be as follows. Consider a signal $h(t) = A \exp(i2\pi f_0 t + \phi)$. The Fourier transform spectrum of this signal at the frequency f_0 would return the amplitude of A and the phase constant ϕ . In order to carry this understanding of phase into the realm of wavelet theory, time-frequency representations and local spectra, it would require a transform that would return a voice (function of time) for the frequency f_0 that had a constant amplitude A and a constant phase ϕ . Such is the case with the S-transform, and thus it can be described as a generalization of the Fourier transform to the case of nonstationary signals.

In summary the S-transform has the following unique properties. It uniquely combines frequency dependent resolution with absolutely reference phase, and therefore the time average of the S-transform equals the Fourier spectrum. It simultaneously estimates the local amplitude spectrum and the local phase spectrum, whereas a wavelet approach is only capable of probing the local amplitude/power spectrum. It independently probes the positive frequency spectrum and the negative frequency spectrum, whereas many wavelet approaches are incapable of being applied to a complex time series. It is sampled at the discrete Fourier transform frequencies unlike the CWT where the sampling is arbitrary. Because of the absolutely referenced phase of the S-transform, it is possible to define a channel instantaneous frequency function for each voice [19]. It is possible to perform a localized cross spectral analysis as shown in [20,21]

where transient wavelike motions were detected in the time series measured by spaced receivers (physically separated by a known distance). By analyzing phase shifts in the S-transforms of the two time series, the phase speed of the wavelike motion was deduced. Such a straightforward analysis is not possible with a wavelet type approach.

The goal of this paper is to define a discrete orthonormal S-transform (DOST) representation that encompasses these properties.

3. The basis functions of the discrete orthonormal S-transform (DOST)

There are several reasons to desire an orthonormal time-frequency version of the S-transform. An orthonormal transformation takes an N -point time series to an N -point time-frequency representation, thus achieving the maximum efficiency of representation. Also, each point of the result is linearly independent from any other point. The transformation matrix (taking the time series to the DOST representation) is orthogonal, meaning that the inverse matrix is equal to the complex conjugate transpose. By being an orthonormal transformation, the vector norm is preserved. Thus a Parseval theorem applies, stating that the norm of the time series equals the norm of the DOST. An orthonormal transform is referred to as an energy preserving transform.

The efficient representation of the S-transform can be defined as the inner products between a time series $h[kT]$ and the basis functions defined as a function of $[kT]$, with the parameters ν (a frequency variable indicative of the center of a frequency band and analogous to the “voice” of the wavelet transform), β (indicating the width of the frequency band), and τ (a time variable indicating the time localization).

$$S\{h[kT]\} = S\left(\tau T, \frac{\nu}{NT}\right) = \sum_{k=0}^{k=N-1} h[kT] S_{[\nu, \beta, \tau]}[kT]. \quad (5)$$

These basis functions $S_{[\nu, \beta, \tau]}[kT]$ for the general case are defined as

$$S_{[\nu, \beta, \tau]}[kT] = \frac{ie^{-i\pi\tau}}{\sqrt{\beta}} \frac{\{e^{-i2\pi(k/N - \tau/\beta)(\nu - \beta/2 - 1/2)} - e^{-i2\pi(k/N - \tau/\beta)(\nu + \beta/2 - 1/2)}\}}{2 \sin[\pi(k/N - \tau/\beta)]}. \quad (6)$$

At this point, the sampling of the time-frequency space has not yet been determined. Rules must be applied to the sampling of the time-frequency space to ensure orthogonality. These rules are as follows:

- Rule 1. $\tau = 0, 1, \dots, \beta - 1$.
- Rule 2. ν and β must be selected such that each Fourier frequency sample is used once and only once.

Implicit in this definition is the phase correction of the S-transform that distinguishes it from the wavelet or filter bank approach. Here the parameters ν, β, τ are integers defined such that the functions do form a basis. For each voice, there are one or more local time samples (τ), this number being equal to β (see Rule 1) thus the wider the frequency resolution (large β), the more samples in time (large τ). This can be seen as a consequence of the uncertainty principle. Examples and methods for determining these parameters are described below.

Distinct from a wavelet function, these basis functions have no vanishing moments (in fact the functions are normalized to unit area). These basis functions are not translations of a single function, and they are not self-similar.

3.1. Orthonormal basis functions with octave sampling

In order to compare the DOST with orthonormal wavelet transforms and with the S-transform, an octave sampling of the time-frequency domain is illustrated. This has the property of progressive resolution that both the S-transform and wavelet transforms share. Octave sampling implies that the voice bandwidth doubles for each increasing voice (as sampling allows).

By imposing specific rules on the basis functions (here octave sampling) it implies a strict definition for ν and β . By introducing a new variable p which corresponds to the octave number $p = 0, 1, 2, \dots, \log_2(N) - 1$ one can define all the parameters (ν, β, τ) of equation 6 in terms of p as follows:

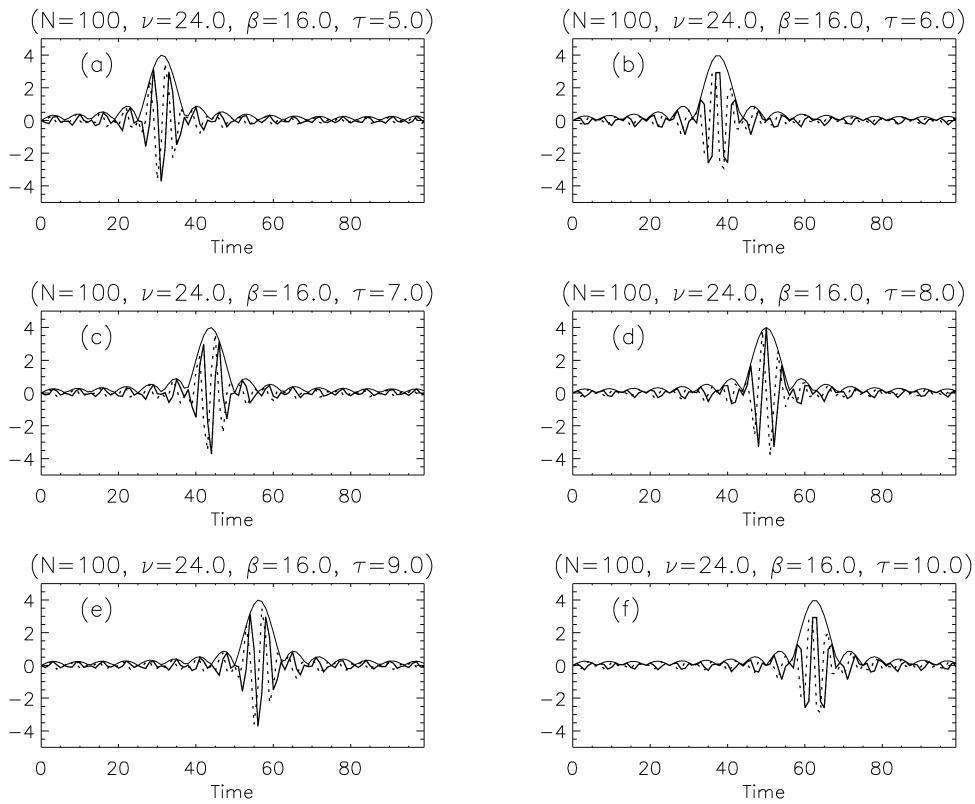


Fig. 1. Selected DOST basis functions of increasing time with octave sampling for $N = 100$. Note that the DOST functions are not translates of a single function. Each plot is a basis function with a different time index τ for the same voice frequency. Note that the time index parameter τ is scaled by β/N . Thus $\tau = 10$ corresponds to a shift of $t = \tau N/\beta = 10 \times 100/16 = 62.5$ which is where the center of that basis function (f) occurs. The solid line is the real component, the dotted line is the imaginary part, and the light grey trace is the amplitude.

For $p > 1$, we have:

$$p = 2, \dots, \log_2(N) - 1, \quad (7)$$

$$\nu = 2^{(p-1)} + 2^{(p-2)}, \quad (8)$$

$$\beta = 2^{(p-1)}, \quad (9)$$

$$\tau = 0, 1, \dots, 2^{(p-1)} - 1. \quad (10)$$

For the case $p = 0$, then $\nu = 0$, $\beta = 1$, and $\tau = 0$. Also when $p = 1$, then $\nu = 1$, $\beta = 1$, and $\tau = 0$.

Thus the DOST basis functions for octave sampling of a time series is given as follows (by application of (7)–(10) into (6)):

$$S_{[p,\tau]}[kT] = \frac{i e^{-i\pi\tau}}{\sqrt{2^{(p-1)/2}}} \frac{\{e^{-i2\pi(k/N - \tau/2^{p-1})(2^{p-1}-1/2)} - e^{-i2\pi(k/N - \tau/2^{p-1})(2^p-1/2)}\}}{2 \sin[\pi(k/N - \tau/2^{p-1})]}. \quad (11)$$

The application of the DOST with octave sampling for the case where $N = 100$ is illustrated in Figs. 1 and 2. In Fig. 1, octave sampled DOST basis functions are shown for a specific DOST voice ($p = 5$ and therefore $\nu = 24$ and bandwidth $\beta = 16$), for six different time samples ($\tau = 5, \dots, 10$). These functions are not simple translations, as is most evident in a comparison of Figs. 1d and 1f.

Figure 2 also shows six octave sampled DOST basis functions, but now the order p varies (as does voice frequency ν and the bandwidth β). The time sample τ is chosen simply to center the function. Again the lack of the self-similar property is evident. No basis function shown in Fig. 2 is a dilation of any other basis function.

While octave partitioning has the familiar frequency dependent resolution seen in the S-transform and wavelets, the formalism defined in Eq. (6) allows for any arbitrary sampling of the time-frequency space. Often the resolution

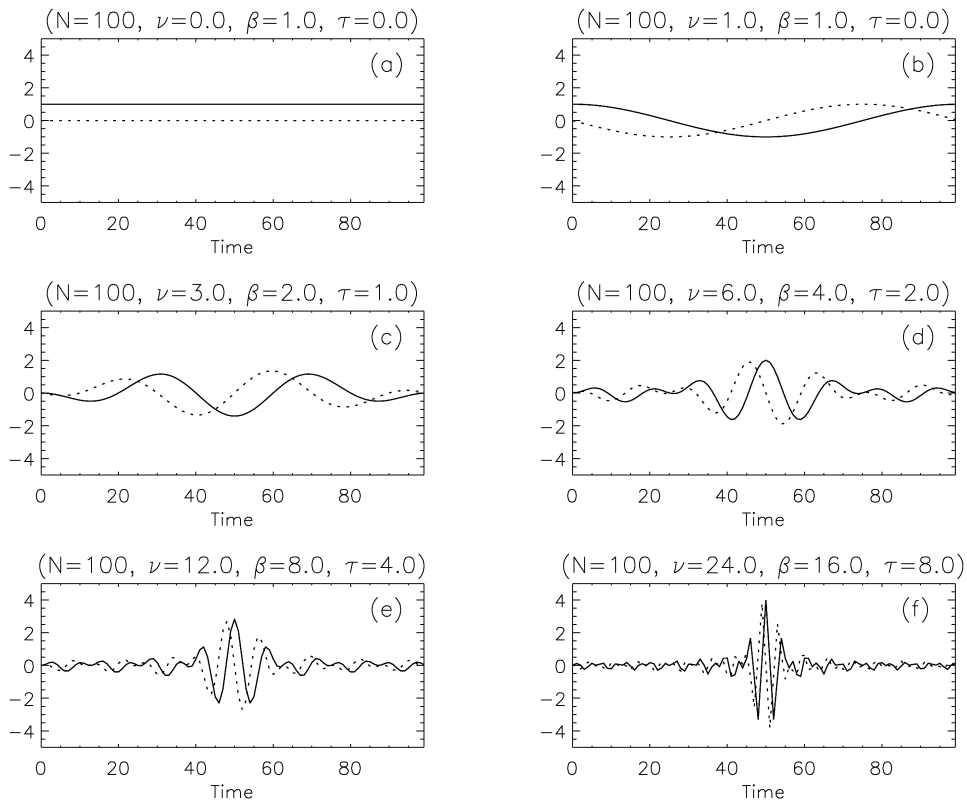


Fig. 2. Selected DOST basis functions of increasing voice frequency with octave sampling for $N = 100$. Each plot shows a basis function (solid line = real, dotted line = imaginary) from a different frequency range (i.e., different voice).

of the time-frequency space is dependent on the source signal; a transient signal will have a broad spectral signature regardless of sampling. Also, while it may be inferred that the above algorithm assumes that the number of points N is a power of two, that is not a strict requirement for octave sampling (as shown in Figs. 1 and 2). The octave sampling scheme can be applied to any arbitrary length time series by starting at the Nyquist frequency with

$$\beta = \text{truncate}(\text{float}(\text{truncate}(N/2) + 1)/2) \quad (12)$$

and for each lower frequency partition, reducing the bandwidth as follows:

$$\beta_{\text{new}} = \text{truncate}(\text{float}(\beta + 1)/2). \quad (13)$$

For example, the octave sampling for $N = 31$ is $\{1, 1, 2, 4, 8, 8, 4, 2, 1\}$ where the first element is the DC level, then the next four β are the positive frequencies (increasing), and the last four β are the negative frequencies (decreasing).

3.2. Derivation of the basis functions

By extending filter bank theory, in combination with the unique phase correction of the S-transform, the time domain basis functions for the S-transform are developed. The novel idea is to create a new orthonormal basis for a time-frequency representation by taking linear combinations of the original Fourier basis functions in band limited subspaces. Within a frequency band (i.e., a particular voice), several orthogonal basis functions are formed by a linear combination of the Fourier basis functions in that frequency band. There are β components in this operation. Thus β basis functions can be derived by applying the appropriate phase functions to the components (where each of the β basis functions is indexed by $\tau = 0, 1, \dots, \beta - 1$). The key to creating orthogonal functions is the careful selection of the frequency shift applied to the Fourier basis functions. This action is the analog of the “phase correction” of the S-transform. This is where the absolutely referenced phase information originates, and it is what distinguishes these basis functions from wavelets (i.e., they are not self-similar as shown in Figs. 1 and 2 and also below in Fig. 4). It is

also the inclusion of this additional operation that is distinct from a simple filtering operation, as indicated below in Figs. 8 and 9. Note, the voice shown in Fig. 9b cannot be created by a filtering operation of the time series in Fig. 8a.

The basis functions can be derived by starting with a partitioning of the spectrum (a simple restricted sum of complex-valued Fourier basis functions) defined in the time domain (function of $[kT]$), centered at frequency ν with a bandwidth of β and applying the appropriate phase and frequency shifts:

$$S(kT)_{[\nu, \beta, \tau]} = \frac{1}{\sqrt{\beta}} \sum_{f=\nu-\beta/2}^{f=\nu+\beta/2-1} \exp\left(i2\pi \frac{k}{N} f\right) \exp\left(-i2\pi \frac{\tau}{\beta} f\right) \exp\left(-i2\pi \frac{N}{2NT} \tau T\right), \quad (14)$$

where $1/\sqrt{\beta}$ is a normalization factor to insure orthonormality of the basis functions.

Thus the basis function for the discrete orthonormal S-transform (DOST) of voice frequency ν , bandwidth β , and time index τ can be written as

$$S(kT)_{[\nu, \beta, \tau]} = \frac{e^{-i\pi\tau}}{\sqrt{\beta}} \sum_{f=\nu-\beta/2}^{f=\nu+\beta/2-1} \exp\left(i2\pi \left(\frac{k}{N} - \frac{\tau}{\beta}\right) f\right). \quad (15)$$

Application of the identity

$$c + cx + \cdots + cx^{n-1} = \frac{c(1-x^n)}{1-x} \quad (16)$$

to Eq. (15) leads to Eq. (6). As can be seen in Eq. (15), the function has no poles (β is always greater than zero). In Eq. (6), there is an apparent pole where the denominator goes to zero (where $k/N \rightarrow \tau/\beta$), but application of L'Hopital's rule shows that the limit of the basis function as $k/N \rightarrow \tau/\beta$ is well behaved and equal to:

$$\lim_{k/N \rightarrow \tau/\beta} S(kT)_{[\nu, \beta, \tau]} = \sqrt{\beta} e^{i\pi\tau}. \quad (17)$$

One advantage of this method is that one can directly calculate any voice, without having to iterate through a series of intermediate steps. Also, there is no filter design involved nor any upsampling or downsampling algorithms required. Another advantage is that it allows one to directly apply the ideas of power spectrum estimation, such as applying windows and apodizing functions, to the analysis of the local spectrum of a time series.

Note that in a departure from filter bank theory, the sum is centered on the voice frequency ν . In other words, a frequency translation has been applied. The operation of calculating the inner product of a time series with this basis function not equivalent to a simple filtering operation (in the asymptotically simple case of the time series consisting of an oscillating sinusoid, the resulting voice will be a constant, in amplitude and phase, for each time sample). This frequency shift is vital when the characteristic of absolutely referenced phase information (and cross local spectrum analysis and generalized instantaneous frequency) is described, and is the distinguishing difference between the S-transform approach, and the wavelet/filter bank approach. This shift in frequency is the key feature of the original S-transform. In Eq. (11) of [19], the “ n ” voice $S[jT, n/NT]$ has the same frequency translation applied by the shift of the spectrum $H[(m+n)/NT]$ by “ n ” which centers the spectrum around the n frequency.

$$S\left[jT, \frac{n}{NT}\right] = \sum_{m=0}^{N-1} H\left[\frac{m+n}{NT}\right] e^{-2\pi^2 m^2/n^2} e^{i2\pi m j/N}, \quad n \neq 0. \quad (18)$$

It is precisely this property of the basis functions that provides absolutely referenced phase information, and it is also this property that implies that the basis functions are not self-similar.

It is easy to show that the basis functions are indeed orthonormal and have compact support in frequency. They are not compactly supported in time, but they are local. The property of “compact support” refers to a particular transform. An orthonormal wavelet does not have compact support under a Fourier transform. By the uncertainty principle, one cannot have a compactly supported function which has a compact Fourier transform. These basis functions are compact in frequency, and also local in time, while maintaining orthogonality. All that is required is that the ν and β values are chosen such that the bandwidths do not overlap, and that all discrete frequencies are sampled.

The utility of these basis functions is that they create a road map for one to overlap bandwidths and oversample in time in an arbitrary (perhaps data adaptive) manner to achieve any desired sampling of the time-frequency space.

3.3. Inverse transform

The inverse of the transform can be quickly calculated as a straightforward reconstruction of the Fourier spectrum from the DOST voices (by application of the forward discrete Fourier transform to each voice), reversing the spectral partitioning process of Eq. (15). The time series is then recovered by use of the inverse Fourier transform of this spectrum.

4. The relationship between the wavelet transform and the DOST

The fundamental difference between the wavelet transform and the DOST is in the absolutely referenced phase information. The phase of a voice of the DOST has the same meaning as the phase of a component of the Fourier transform. This is in contrast to the meaning of the phase of a wavelet transform, including the complex Daubechies wavelets for which there is no direct equivalence as to the meaning of phase.

The fundamental idea of a wavelet transform is the decomposition of a signal into deformations of a mother wavelet. In the analysis of geophysical data, often the local characteristics of the phase of the signal can lead to great insight. In order to resolve the local phase properties, it is necessary to break the analyzing wavelets from the mother wavelet, and impart an absolutely referenced phase to the basis functions. This break separates the time-localizing windowing function, from the frequency-localizing modulation. As the time-localizing window translates in time, it is essential for this formalism that the frequency modulation does not translate in time, which then forbids the possibility of self-similar basis functions. One can either have a self-similar basis (wavelets), or an absolutely referenced phase basis (DOST). Another departure from the requirements of the definition of the wavelets is that the mean of a DOST basis function is not zero, rather it is normalized to one.

5. Comparison with complex wavelets

One desirable feature of these basis functions is that it is fully complex valued (as is the discrete Fourier transform) and therefore one can localize the positive frequency spectrum independently from the negative frequency spectrum. More recently there have been investigations into complex-valued orthogonal wavelets (Lina [13], Kingsbury [14–16], Zhang [45], and Selesnick [46]). While they have found utility in the phase of the complex wavelets there is no direct equivalence between the Fourier transform and wavelets as to the meaning of phase. This is in direct contrast to the DOST functions where there is a very clear relationship between the Fourier transform phase, and the DOST phase. As a result, the DOST has many desirable characteristics such as a definition of generalized instantaneous frequency, localized cross spectral analysis, and a direct collapse over the time variable into the Fourier transform spectrum. All of these characteristics are unique to the DOST in the orthogonal case, and to the S-transform in the redundant representation case.

5.1. Dual tree complex wavelets

In Fig. 3 the dual tree complex wavelets [16] at several scales are shown, along with the accompanying amplitude spectra. The spectral response of these wavelets is quite broad. The dual tree transform is expansive ($n \rightarrow 2n$), while the DOST is an orthogonal decomposition ($n \rightarrow n$). While a better representation of local spectral responses is desired for geophysical applications, the dual tree wavelets have found success in image processing applications [14,15].

5.2. Complex Shannon wavelets

The wavelet family most similar to the DOST is the complex Shannon wavelet

$$\text{Sh}(t) = (F_b^{0.5}) (\text{sinc}(F_b t) \exp(i 2\pi F_c t)), \quad (19)$$

where F_b is the bandwidth and F_c is the center frequency.

To better compare the Shannon wavelets with the Dost basis functions, they can be modified with the following substitutions $t \rightarrow k/N - \tau/\beta$, $F_c \rightarrow \nu$, and $F_b \rightarrow \beta$ also:

$$\text{Sh}(t) = (\beta^{0.5}) \frac{\sin(\beta(k/N - \tau/\beta))}{\nu(k/N - \tau/\beta)} \exp(i 2\pi \nu(k/N - \tau/\beta)). \quad (20)$$

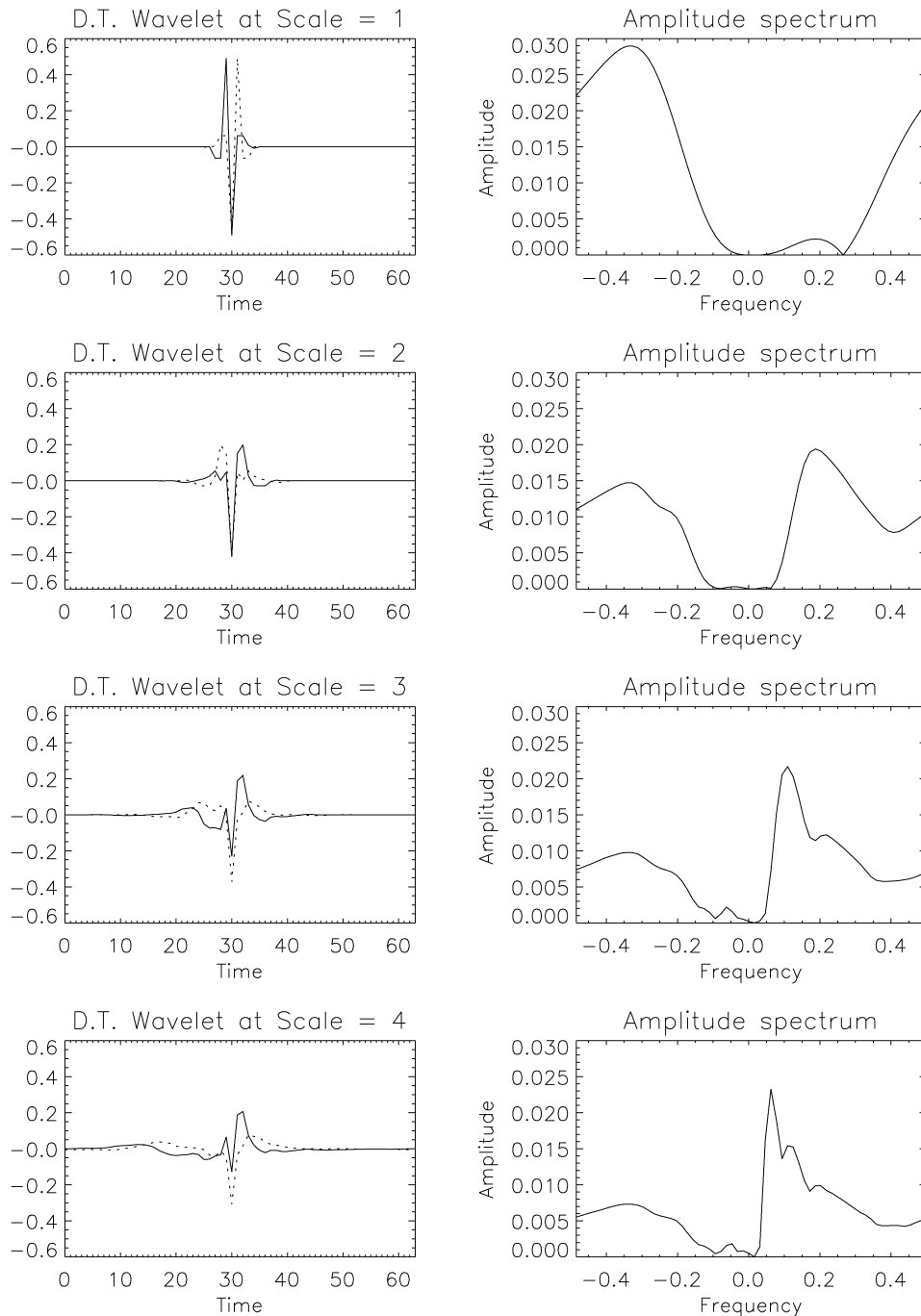


Fig. 3. Several scales of the dual tree complex wavelets for $N = 64$ (left column). The right column shows the corresponding amplitude spectra for the wavelets. The spectral response of the wavelets is broad.

Figure 4 is fundamental to understanding the “phase corrected nature” of the DOST (as well as the S-transform). For comparison the complex Shannon wavelet has been normalized. In the wavelet transform, a basis function for a different time τ is translated (this is an aspect of the self-similar property) as is shown in the traces marked “Shannon.” For the DOST, the basis function is not translated (and therefore the basis functions are not self-similar); only the envelope has been translated to localize a later time. In Fig. 4, arrows point to the center of the complex Shannon wavelet for each of the translations. This point is the local point to which the phase of the wavelet is referenced, and it

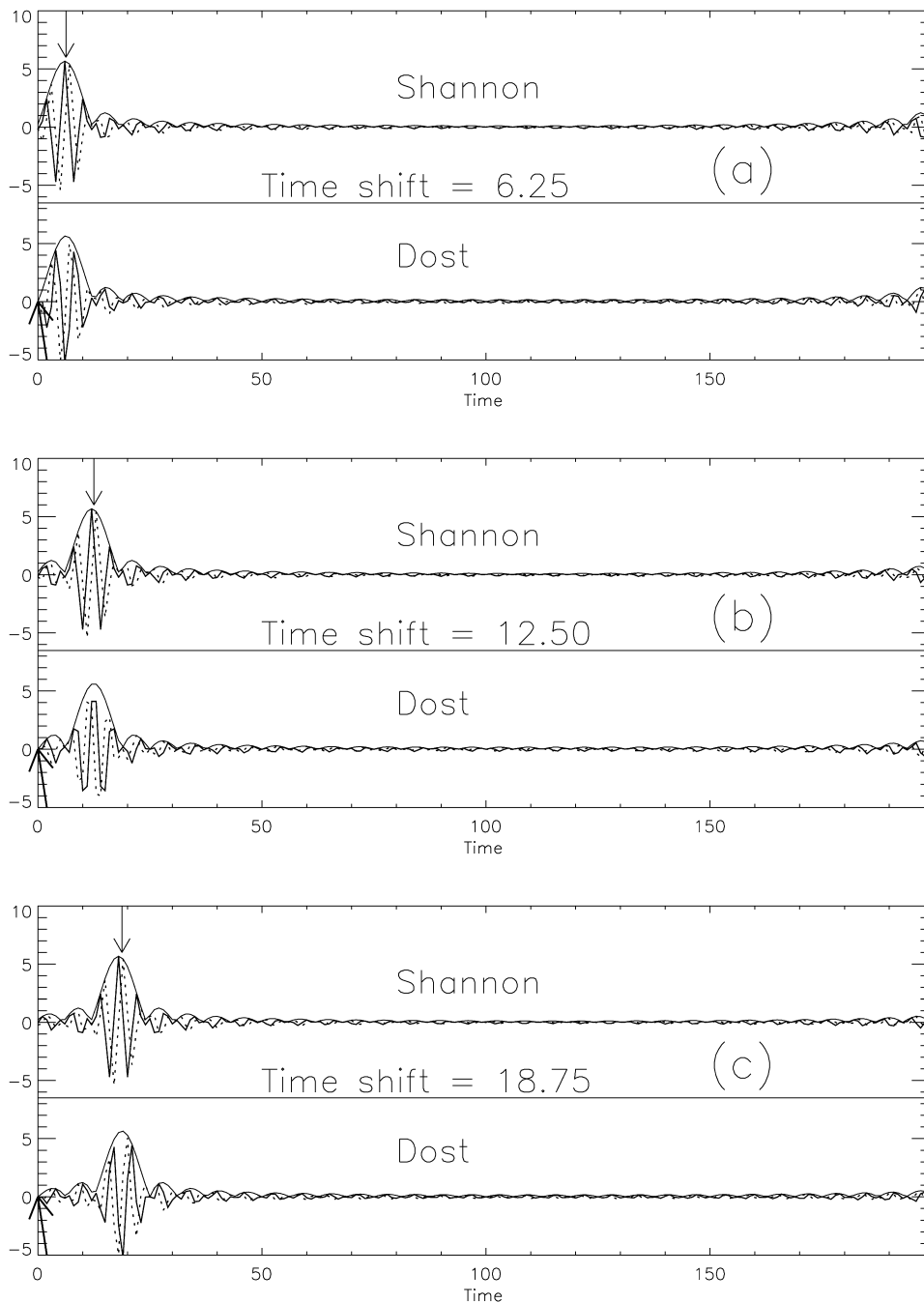


Fig. 4. Illustration of the complex Shannon wavelet (upper trace), with the DOST basis functions (lower trace). For one particular voice (with fixed bandwidth and center frequency) several time samples are shown for both functions. The complex Shannon wavelets shown are not orthogonal to each other. The DOST functions for this voice are not self-similar, but maintain orthogonality.

obviously translates as the wavelet translates. For the DOST, the reference point for the phase remains fixed at $t = 0$ (hence the absolutely referenced phase) for the different times. Thus the crests and troughs of the imaginary and real components are fixed, as the envelope of the DOST basis function translates.

As is well known from Fourier theory, the translation in time of the wavelet (or any function) corresponds to a phase modulation in the Fourier spectral domain. Thus wavelet voices have a phase modulation applied to them in the spectral domain. This is not true for the DOST as there is no direct translation applied to the basis function to produce

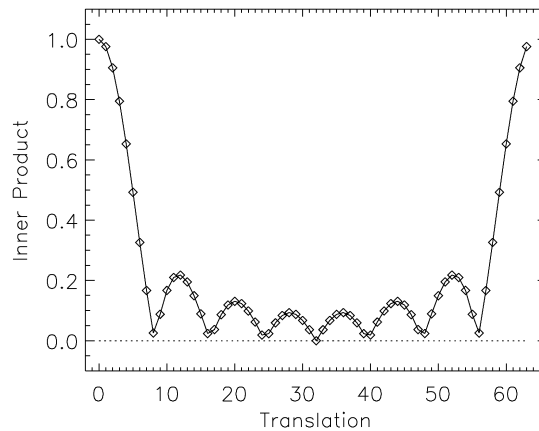


Fig. 5. Calculation of the inner product of the complex Shannon wavelet for each of the 64 translations. Only one translation (length/2) is orthogonal, and thus the complex Shannon wavelets cannot define a basis for the vector subspace of this voice.

other basis functions. It is this characteristic of self-similarity that force this phase modulation (and therefore the lack of absolutely referenced phase) on all wavelet transforms.

In Fig. 4, the DOST functions at these time samples are orthogonal, whereas the Shannon wavelets for these time samples are not orthogonal. To further illustrate this, Fig. 5 shows the inner product of the complex Shannon wavelet (at $t = 0$) with the translated wavelet for each possible translation ($t = 1, 2, \dots, 63$). In only the one case is the inner product zero (at $t = 64/2$), hence the complex Shannon wavelets cannot be used to create a basis for this bandlimited subspace. In stark contrast, the DOST functions do provide such a basis, not only of the whole spectral space but also in each band-limited subspace, due to the correct phase of the defining functions.

6. Applications: Localized cross spectral analysis

As an illustration of the efficient representation that these basis functions afford, they are applied to a one million point ($N = 1,048,576$) time series in Fig. 6. Due to the efficient time-frequency representation of the orthogonal transform, the time-frequency representation also has only 1,048,576 points. This two-dimensional representation shown here is created by making a shaded polygon for each of these 1,048,576 points, with its boundaries indicating the time and frequency resolution. Thus there are 1,048,576 polygons spanning the time-frequency space. An equivalent S-transform of this time series would require the calculation of an unwieldy 10^{12} complex valued points (requiring some 8 terabytes of storage space).

Because of the phase characteristics of the DOST, one can determine the phase shifts between each component of two time series containing transient sinusoidal signals, as is illustrated in the next example. Here a one million point ($N = 1,048,576$) time series is created with three transient sinusoidal components. Each of these three components has its own phase constant, and all components have an amplitude of 1. The details of the first time series are listed in Table 1. As an illustration of the localized cross spectral analysis, a second similar time series ($N = 1,048,576$) is also created with the three transient components, with each component having a different phase constant. The details of the second time series are listed in Table 2.

The localized cross spectral function of a time series $h_1[kT]$ with a DOST $S_1(\tau, \nu)$ and a time series $h_2[kT]$ with a DOST $S_2(\tau, \nu)$ is defined as

$$\text{LCSF}(\tau, \nu) = S_1(\tau, \nu) S_2^*(\tau, \nu) = A(\tau, \nu) e^{i\Phi(\tau, \nu)}, \quad (21)$$

where the asterisk denotes complex conjugation. The meaning of this local cross spectral function is that given a particular voice ν and time index τ , there is a phase shift given by $\Phi(\tau, \nu)$ and a cross spectral amplitude given by $A(\tau, \nu)$. With the local spectral representation that DOST provides, it is possible to simultaneously measure the phase shifts of these three transient components, as is shown in Fig. 6. This cannot be done with a wavelet approach.

Figure 6a shows the amplitude image of the DOST of the first time series (Table 1) for the positive frequencies. Each polygon of the image corresponds to one sample of the DOST, which accounts for the blocky appearance. The

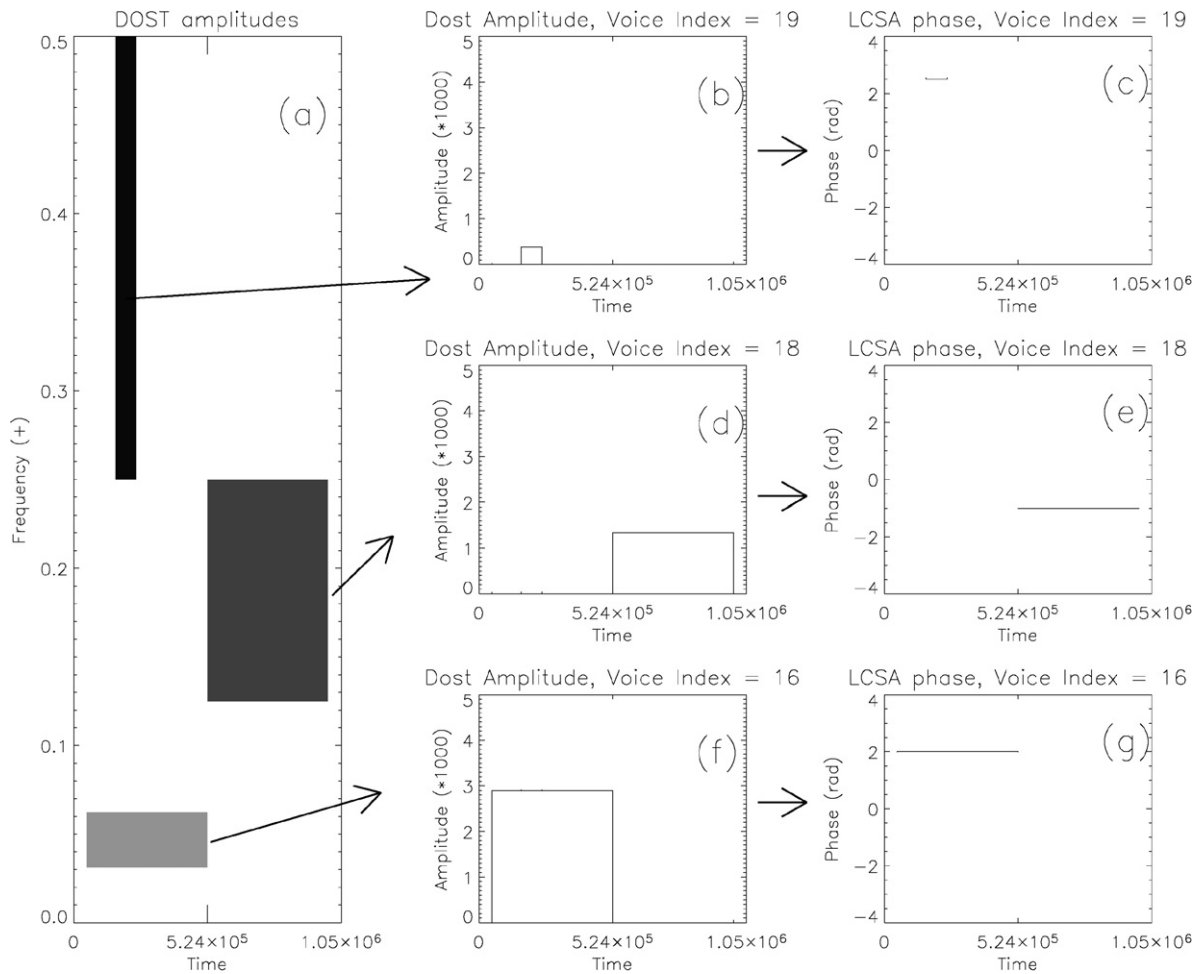


Fig. 6. Illustration of the utility of LCSA. Two very large 1,048,576 point ($t = 0, 1, \dots, 1,048,575$) time series (each containing 3 transient components each with a phase difference). The time series is not shown.

Table 1

The parameters for the first time series in the example

Component	Frequency	Phase constant	Start time	Stop time
High frequency	425,984/1,048,576	0	163,840	245,760
Mid frequency	204,800/1,048,576	0	524,288	998,576
Low frequency	49,152/1,048,576	0	50,000	524,287

Table 2

The parameters for the second time series in the example

Component	Frequency	Phase constant	Start time	Stop time
High frequency	425,984/1,048,576	2.5	163,840	245,760
Mid frequency	204,800/1,048,576	-1	524,288	998,576
Low frequency	49,152/1,048,576	2	50,000	524,287

middle column shows the amplitude for a voice of the DOST (i.e., a single row of the DOST image in Fig. 6a). Figures 6c, 6e, and 6g show the LCSA phase function $\Phi(\tau, v_0)$ of Eq. (21) for three different voices (corresponding to the three different transient components of the time series). For Fig. 6c the phase of the cross DOST voice ($p = 19$)

is equal to 2.5 radians as expected, for Fig. 6e the phase of the cross DOST voice ($p = 18$) is equal to -1 radians, and for Fig. 6g the phase of the cross DOST voice ($p = 16$) is equal to 2 radians.

7. Applications: Channel instantaneous frequency

The use of instantaneous frequency in a local spectral representation was introduced for the S-transform [19], and independently was employed by Nelson [47] to study nonstationary multicomponent FM signals. Gardner and Magnasco [48] also used band-pass filtered instantaneous frequency analysis for the analysis of sound.

As with the S-transform, the DOST provides an extension of instantaneous frequency to broadband signals. This is the first time a channel instantaneous frequency analysis has been performed with an orthogonal time-frequency transform. The voice for a particular frequency ν_1 can be written as

$$S(\tau, \nu_1) = A(\tau, \nu_1) e^{i\Phi(\tau, \nu_1)}. \quad (22)$$

One may use the phase in Eq. (22) to determine the “local” instantaneous frequency similar to that defined by Bracewell [49].

$$\text{GIF}(\tau, \nu_1) = \frac{1}{2\pi} \frac{\partial}{\partial \tau} \{2\pi \nu_1 \tau + \Phi(\tau, \nu_1)\}. \quad (23)$$

Thus the absolutely referenced phase information leads to a generalization of the instantaneous frequency to broadband signals.

In Fig. 7, the first time series of the above example (Fig. 6) is used (the details are in Table 1). The frequency of the two higher frequency components is different from the voice frequencies of the octave partitioned DOST. For the low frequency component, the signal has the same frequency as the voice of the DOST.

For each case, the generalized instantaneous frequency (GIF) is calculated from a single voice (corresponding to three transient components of the time series). The phase of this DOST voice is calculated and a simple phase unrolling algorithm is applied to it (where jumps of greater than π between successive points are replaced by jumps less than π by adding an appropriate factor of 2π to the phase). The GIF is calculated using IDL’s `deriv()` function which utilizes a 3-point Lagrangian interpolation. Thus the GIF in Figs. 7b, 7d, and 7f were calculated as

$$\text{GIF}(\nu, \tau) = \text{deriv}(\text{phaseunroll}(\arg(\text{voice}))) \frac{\text{length}}{2\pi\beta}, \quad (24)$$

where $\arg() = \text{atan}(\text{imaginary}(\text{voice}), \text{real}(\text{voice}))$, and length is the number of points in the time series.

In Fig. 7b, the slope of the phase in Fig. 7a is shown, and is equal to a frequency shift of $32,768/1,048,576$ indicating (as expected) the signal’s high frequency component $425,984/1,048,576$ (Table 1). In Fig. 7c, the phase of the DOST for the 18th voice ($\nu = 196,608/1,048,576$) is shown. Fig. 7d shows the slope of the phase in Fig. 7c which is equal to frequency shift of $8,192/1,048,576$ giving the signal’s high frequency component $204,800/1,048,576$ (Table 1). Fig. 7e shows the phase of the DOST for the 16th voice ($\nu = 49,152/1,048,576$) and Fig. 7f shows the slope of the phase of Fig. 7e. This is equal to a frequency shift of $0/1,048,576$ giving the signal’s high frequency component $49,152/1,048,576$ (Table 1). The measure of instantaneous frequency is not quantized to the sampled frequencies, and thus can achieve arbitrarily high accuracy. In essence, the application of the generalized instantaneous frequency can be utilized as a local peak-finding algorithm. Again, there are ringing effects and wrap around effects that are familiar in discrete Fourier transform analysis. While this example is shown with a simple sinusoidal signal, the generalized instantaneous frequency is useful in the general case [47,48].

8. Application: Analysis of a complex time series

Spectral analysis of a two component time series has been performed [50] on data such as the horizontal wind field $[u(t), v(t)]$ by constructing a one-dimensional complex-valued time series as follows:

$$U[t] = u[t] + \iota v[t] = A(t) e^{i\omega t}, \quad (25)$$

where $u[t]$ is the zonal wind (positive eastward), $v[t]$ is the meridional wind (positive northward), $\iota = \sqrt{-1}$, ω is the radial frequency of rotation, and $A(t)$ is the amplitude function. The Fourier transform of a complex valued time

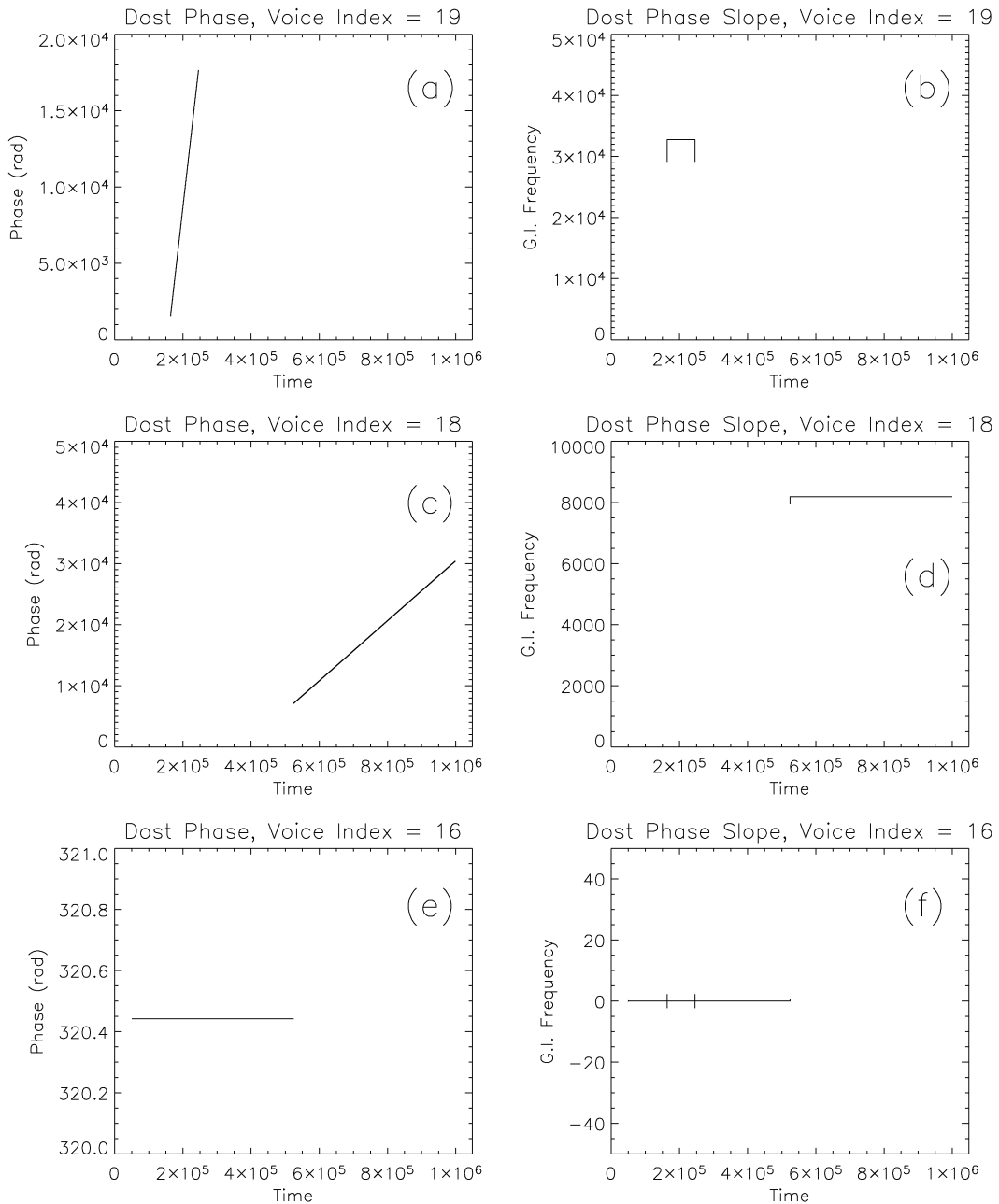


Fig. 7. Illustration of the utility of the generalized instantaneous frequency (GIF). Here the first 1,048,576 point time series of Fig. 6 is analyzed to calculate the GIF for 3 voices. The unrolled phase is only plotted in regions where the amplitude of the DOST is near of the maximum value.

series requires both the positive and negative frequencies to completely characterize the wind. Such “rotary spectral” analysis will decompose the wind motions into circular (clockwise and counter-clockwise) components.

As with the Fourier transform, both the S-transform and the DOST can be applied to a 2 component vector time series such as the horizontal wind field $[u(t), v(t)]$ by constructing a one-dimensional complex valued time series as in Eq. (25). S-transform interpretations for a variety of canonical rotating time series are summarized as follows:

- Circular clockwise implies purely negative frequencies.
- Circular counter-clockwise implies purely positive frequencies.
- Elliptical clockwise implies negative frequency dominant.

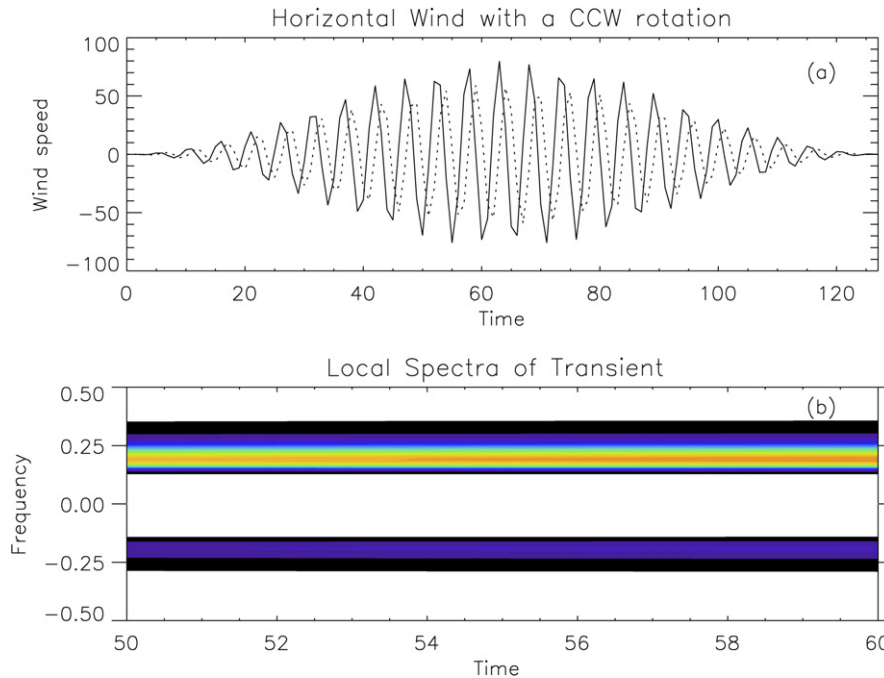


Fig. 8. (a) A two component time series (analogous to a horizontal wind measurement for instance) that initially increases in amplitude, then decreases in amplitude. Additionally, the wind is rotating counter-clockwise. (b) The S-transform amplitude. The changing amplitude can be seen. The counter-clockwise motion can be seen in that the positive frequency component is larger than the negative frequency component.

- Elliptical counter-clockwise implies positive frequency dominant.
- In (de)creasing spiral implies in (de)creasing amplitude.
- Linear implies positive and negative frequency amplitude equal.

By looking at the phase difference between the positive and negative frequencies it is possible to infer complete information of the generalized elliptical motion, including the sense of rotation, the major and minor axes, and ellipticity [51]. A synthetic time series illustrating this effect is shown in Fig. 8. The complex time series depicts a counter-clockwise rotating surface wind vector, with an amplitude that starts small, grows in time, then decays. The time series was constructed in the form of Eq. (25) as follows $U[t] = 80 \cos(2\pi t 24.4/128) + i 60 \sin(2\pi t 24.4/128)$, where $t = 0, 1, \dots, 127$ and the amplitude is varied by multiplying the time series with a hanning window. The amplitude of the S-transform of this time series is shown in Fig. 8b. The white background denotes regions of negligible energy, and filled regions correspond to large amplitudes. The peak amplitude occurs at $f = 24/128$ and $t = 64$. Because this amplitude is larger than its mirror in the negative frequencies ($f = -24/128$ and $t = 64$), this represents a counter-clockwise oscillation. If the negative frequency is larger, then rotation is clockwise. The result of applying the DOST to the same time series is shown in Fig. 9. The DOST amplitudes plotted in time-frequency space are shown in Fig. 9a. Both the S-transform and the DOST have a similar appearance, however the DOST has only 128 points, whereas the S-transform has 16,384 points. The DOST shows the same feature as the S-transform, indicating counter-clockwise rotation of the time series. The voice of the DOST for the positive frequency voice ($v = 5$) is plotted in Fig. 9b. Note that this voice cannot be achieved by a filtering operation of the time series in Fig. 8a.

9. Analogy to power spectral density estimation

There is a strong analogy between DOST analysis and Fourier analysis regarding the identification and estimation of spectral peaks. In the nonwindowed orthogonal form, the DOST will have “ringing effects” and exhibit sidelobe contamination as does the DFT. In order to reduce the variance of a frequency bin (DFT) or of a time-frequency bin (DOST) it is possible to smooth the data (via convolution) in the time domain, or equivalently window the data in the frequency domain. This is recommended especially if one is employing the LCSA or GIF algorithms. As this window

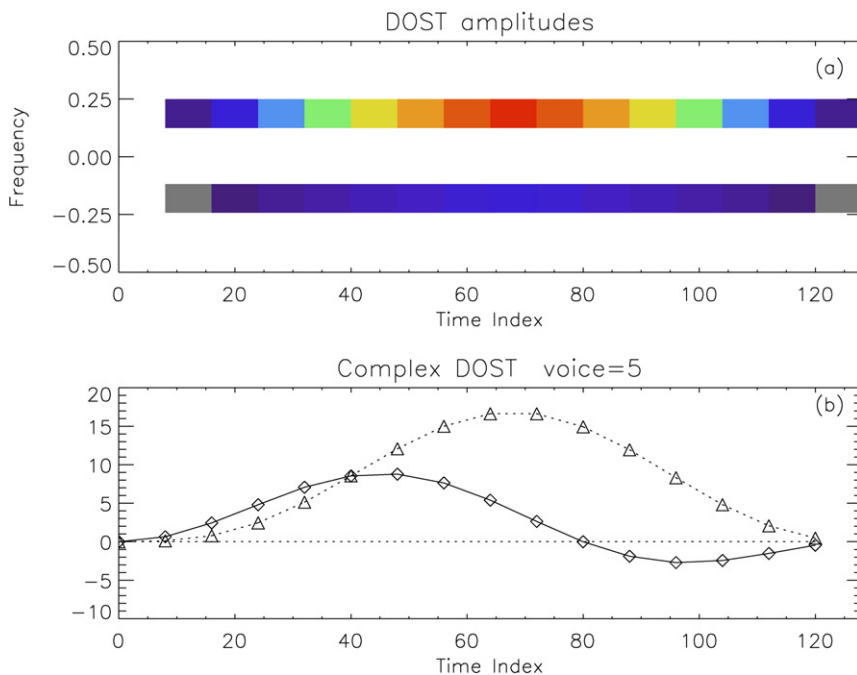


Fig. 9. (a) The amplitude of the DOST plotted in the time-frequency domain (compare to Fig. 8b). Because of the sparse orthogonal nature of the DOST, the image is blocky. It does however show the counter-clockwise rotation of the time series because the positive frequency component is larger than the negative frequency component. (b) The voice of the DOST for the positive frequency voice ($\nu = 5$) showing the response of the complex time series of Fig. 8a. The solid line is the real component and the dotted line is the imaginary component.

is shifted in the spectral domain, it acts as a convolving apodizing function. In the case of the DOST, a window can be applied by multiplying the partitioned spectrum with an appropriate window. Such application may alter the properties of the DOST as it does for the discrete Fourier transform.

For certain purposes it is highly desirable to window and interpolate the DOST time-frequency representation and approach the more ample sampling of the S-transform. Use of windowing and interpolation allows one to build up a smoother and less sparse representation of the time-frequency space, and in the extreme case of “fully redundant” interpolation (i.e., where a time-frequency representation point is calculated for each time sample of the original time series and for each frequency sample of the Fourier spectrum) then one can recreate the action of the original S-transform. The DOST forms the essential skeleton of the time-frequency plane, from which the fully redundant S-transform can be derived. This is shown in Fig. 10. A modulated sinusoid time series is shown in Fig. 10a ($h[t] = \cos(2\pi \times (204 + 6 \times 1024/(1+t)) \cos(2\pi \times 3t/1024))t/1024$, where $t = 0, 1, 1023$). The amplitude DOST is shown in Fig. 10b, and the smooth redundant S-transform amplitude is shown in Fig. 10e, with intermediate steps shown in Figs. 10c and 10d. Thus one can achieve any arbitrary level of compromise between efficient representation (DOST) and a smooth and redundant representation by arbitrary choice of oversampling and windowing. The application of windowing to the DOST is easily done when one realizes that an implicit boxcar window is applied to the spectral partition in the implementation of the DOST algorithm. In fact, any window (or apodizing function) can be applied at that stage. This is called “voice windowing” and is analogous to the “voice Gaussian” of the S-transform [19]. If one applies such a window, and thus reduces one’s sensitivity at the edges of the frequency bands, then it becomes desirable to interpolate and oversample in frequency. When a window is applied to the partitioned spectrum, one should expect this new band to be representative of frequencies $\nu \pm \text{bandwidth of window}$, rather than (in the case of unwindowed DOST) $\nu \pm \beta/2$.

If one wished to interpolate the DOST in time, this can be accomplished by widening the partitions in the frequency domain so that they overlap (thus the partitioned spectrum is then extended to more higher and lower frequencies interpolating the time domain voice). As one goes to full interpolation in time (i.e., a sample at every time sample of the original time series) then there is no partitioning (each spectral partition is the original spectrum). Only the shift of the spectrum (centering the voice frequency ν) and the voice-windowing will differentiate different frequency ranges.

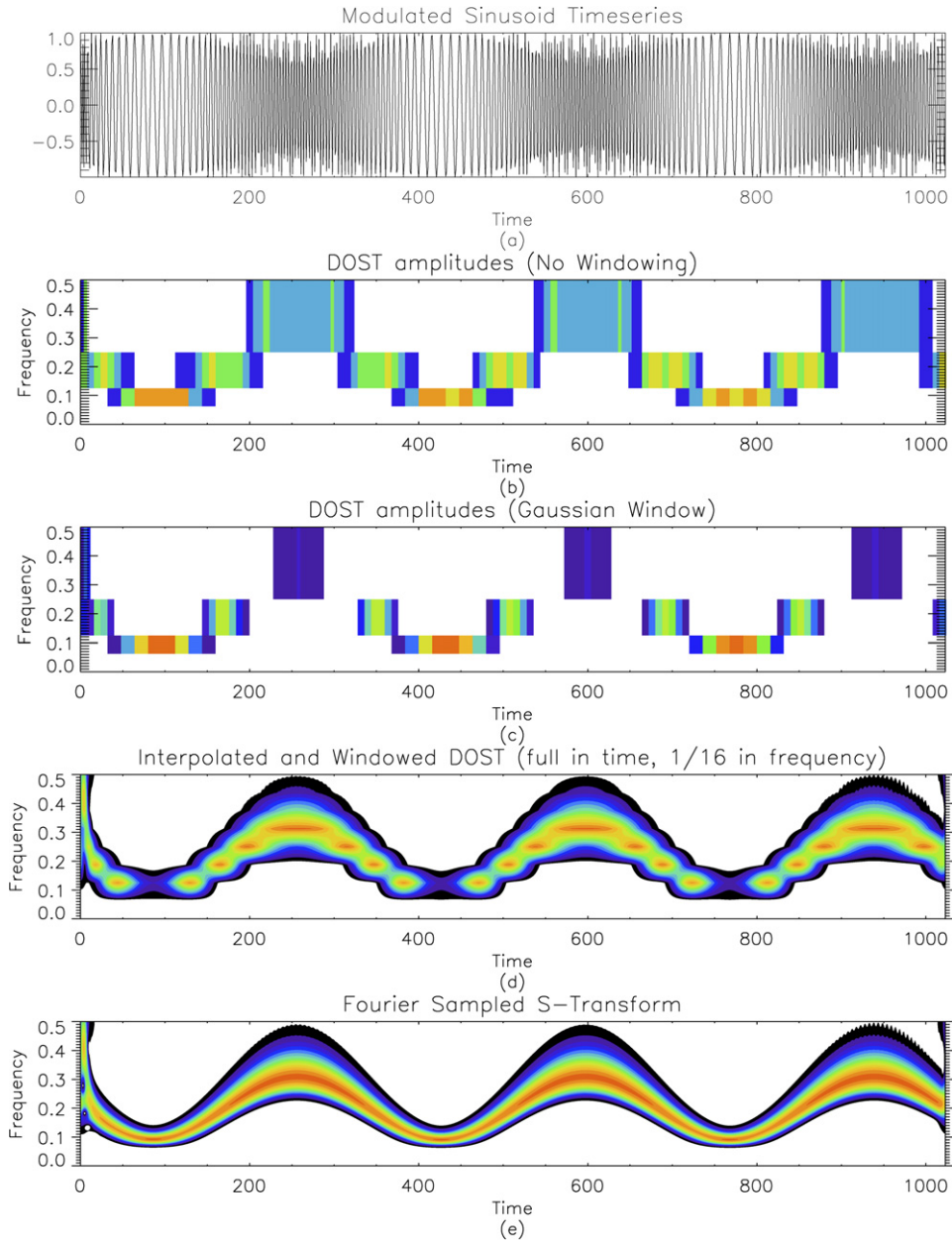


Fig. 10. Illustration of the progression from the efficient DOST time-frequency representation (b) to the redundant S-transform representation (e) by increasingly interpolated (and windowed) DOSTs. Only positive frequencies and the amplitudes are shown.

9.1. The DFT as a special case of the DOST

In Eq. (14), if one allows order p to run from $0, \dots, N-1$, and chooses $v_0 = \beta_0 = 1$ and $\tau_0 = 0$, then the summation runs from $f = 1$ to $f = 1$ and one is left with

$$S(kT)_{[v_0, \beta_0, \tau_0]} = \sum_{f=1}^{f=1} \exp\left(i2\pi \frac{k}{N} f\right) \exp\left(-i2\pi \frac{0}{1} f\right) \exp\left(-i2\pi \frac{N}{2NT} 0T\right) \quad (26)$$

and when applied to the definition of the DOST Eq. (6):

$$S\{h[kT]\} = \sum_{k=0}^{k=N-1} h[kT] \sum_{f=1}^{f=1} \exp\left(i2\pi \frac{k}{N} f\right) \exp\left(-i2\pi \frac{0}{1} f\right) \exp\left(-i2\pi \frac{N}{2NT} 0T\right), \quad (27)$$

or

$$S\{h[kT]\} = \sum_{k=0}^{k=N-1} h[kT] \sum_{f=1}^{f=1} \exp\left(i2\pi \frac{k}{N} f\right), \quad (28)$$

or

$$S\{h[kT]\} = \sum_{k=0}^{k=N-1} h[kT] \exp\left(i2\pi \frac{k}{N}\right), \quad (29)$$

which is the definition of the discrete Fourier transform. This is a unique feature of the DOST in that the discrete Fourier transform can be derived from a local spectral transform.

10. Summary

In this paper, basis functions have been introduced and used to define the discrete orthonormal S-transform. These basis functions are both local in time and compact in frequency. It is this basis that unifies both the continuous wavelets and the orthonormal wavelet analysis and the time-honored Fourier and spectral analysis.

The advantages of an orthonormal representation are many. Orthogonality implies a robust representation, and conservation of “energy” (as described by a Parseval theorem) is important for statistical analysis. Perhaps the most significant advantage is in the efficiency of representation of the time-frequency information. The orthogonal nature of these functions provide an ideal time frequency representation of a time series in terms of its efficiency and compactness. These basis functions have positive frequency components that are orthogonal to their negative frequency counterpart, as does the Fourier transform basis functions. An important characteristic of the DOST basis functions is the absolutely referenced phase information that they provide. This aspect of the basis functions is shared with the S-transform, and creates a number of unique characteristics absent in wavelet decompositions. It allows one to perform localized cross spectral analysis as illustrated in Fig. 6 and it allows one to perform a generalized instantaneous frequency as illustrated in Fig. 7. It allows one to collapse the DOST time-frequency representation directly into the Fourier transform spectrum, and one can derive the discrete Fourier transform as a special case of the DOST. Because of its close ties to the discrete Fourier transform, the DOST also suffers from many of the same drawbacks such as the familiar problems from sampling and finite length, giving rise to implicit periodicity in the time and frequency domains.

The DOST has the following properties:

- Exact analytical definition of a basis for the S-transform (Fig. 10).
- An orthogonal time-frequency transform from which the discrete Fourier transform can be derived as a special case.
- An orthogonal time-frequency representation that collapses over the time variable to exactly give the discrete Fourier transform spectrum.
- Absolutely referenced phase, thus giving meaning to the phase of an orthonormal time-frequency representation (Fig. 6).
- The ability to directly compare the phase of two time series in a “localized cross spectral analysis” (Fig. 6).
- The ability to employ a channel “instantaneous frequency” to each voice of the DOST (Fig. 7).
- A general definition of a time-frequency representation to which one can apply any of the standard windows of power spectrum analysis in order to perform a localized power spectrum analysis (Fig. 10).
- Introduction of the framework to arbitrarily oversample the orthogonal time-frequency representation on all the way to the redundant S-transform, and thus showing how one can perform time-frequency analysis of almost arbitrarily large geophysical data sets with any desired sampling of the time-frequency space (Fig. 10).
- The DOST can be extended in a straightforward method to higher dimensions for applications such as image processing, and volumetric data analysis, as has been done with the S-transform.

Perhaps the most important use of these basis functions is the development of a foundation to create a more efficient sampled S-transform representation of the time-frequency space. From the framework of this orthogonal basis, one can proceed to oversample the time-frequency space at any desired stage from the sparse but compact orthogonal DOST representation to the fully redundant S-transform.

Acknowledgment

This project was supported by National Science Foundation Grant ATM-0227885.

Appendix A. Explicit derivation of the DOST basis functions

Starting with the basis function for the discrete orthonormal S-transform (DOST) of voice frequency ν , bandwidth β , and time index τ from Eq. (15):

$$S(kT)_{[\nu, \beta, \tau]} = \frac{e^{-i\pi\tau}}{\sqrt{\beta}} \sum_{f=\nu-\beta/2}^{f=\nu+\beta/2-1} \exp\left(i2\pi\left(\frac{k}{N} - \frac{\tau}{\beta}\right)f\right). \quad (\text{A.1})$$

Making the variable substitutions:

$$a = 2\pi(k/N - \tau/\beta), \quad (\text{A.2})$$

$$b = \nu - \beta/2, \quad (\text{A.3})$$

then one has on the right hand side

$$\frac{e^{-i\pi\tau}}{\sqrt{\beta}} \sum_{f=b}^{f=b+\beta-1} \exp(iaf) \quad (\text{A.4})$$

or

$$\frac{e^{-i\pi\tau}}{\sqrt{\beta}} [e^{iab} + e^{ia(b+1)} + \dots + e^{ia(b+\beta-1)}]. \quad (\text{A.5})$$

By application of the identity

$$c + cx + \dots + cx^{n-1} = \frac{c(1-x^n)}{1-x}, \quad (\text{A.6})$$

where $c = e^{iab}$ and $x = e^{ia}$ this is equivalent to

$$\frac{e^{-i\pi\tau}}{\sqrt{\beta}} \left[\frac{e^{iab}(1 - e^{ia\beta})}{1 - e^{ia}} \right] \quad (\text{A.7})$$

multiplying top and bottom by $e^{-ia/2}$

$$\frac{e^{-i\pi\tau}}{\sqrt{\beta}} \left[\frac{e^{-ia/2}e^{iab}(1 - e^{ia\beta})}{e^{-ia/2} - e^{ia/2}} \right] \quad (\text{A.8})$$

leads to

$$\frac{e^{-i\pi\tau}}{\sqrt{\beta}} \left[\frac{e^{ia(b-1/2)}(1 - e^{ia\beta})}{-2i \sin(a/2)} \right] \quad (\text{A.9})$$

or

$$\frac{e^{-i\pi\tau}}{\sqrt{\beta}} \left[i \frac{e^{ia(b-1/2)} - e^{ia(b-1/2+\beta)}}{2 \sin(a/2)} \right]. \quad (\text{A.10})$$

Replacing the values of a and b with τ , β , and ν leads to

$$S_{[\nu, \beta, \tau]}[kT] = \frac{ie^{-i\pi\tau}}{\sqrt{\beta}} \frac{\{e^{-i2\pi(k/N - \tau/\beta)(\nu - \beta/2 - 1/2)} - e^{-i2\pi(k/N - \tau/\beta)(\nu + \beta/2 - 1/2)}\}}{2 \sin[\pi(k/N - \tau/\beta)]}. \quad (\text{A.11})$$

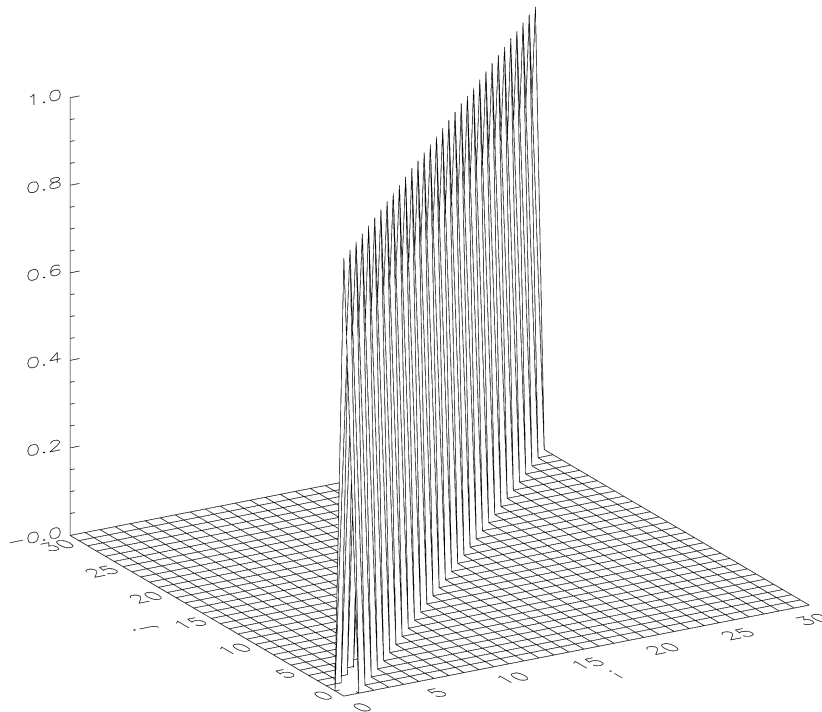


Fig. 11. Matrix showing the dot product of each octave sampled DOST basis function i with every other basis function j . The result is zero if $i \neq j$, and one for $i = j$. Here $N = 64$ and only the positive frequency components are shown.

Appendix B. Orthogonality of basis functions

One can verify the orthogonal nature of the basis functions by calculating the dot product of a basis function i with every other basis function j . This is done in Fig. 11, which shows that the basis functions (with octave sampling) are orthogonal (i.e., $u_i \cdot u_j = \delta_{i,j}$).

Appendix C. Implementation of the DOST

Here is source code written in interactive data language (IDL) for the calculation of the octave sampled basis functions:

```
function basis_definition, N,order,tau,v=v,b=b

if (order ge 2) AND (order le alog(N)/alog(2)-1) then begin
  v = (float(2^(order-1) + 2^(order-2))) ; voice nu
  b = float(2^(order-1)) ; bandwidth beta
  k = findgen(N) ; time index for the time series
  imag = complex(0,-1)
  s = dcomplexarr(N)

  factor=\exp(imag*!dpi*tau)/sqrt(b)
  firstterm=\exp(imag*2*!dpi*(float(k)/N-float(tau)/B)*(v-float(b)/2.-0.5))
  ; second term
  secondterm=\exp(imag*2*!dpi*(float(k)/N-float(tau)/B)*(v+float(b)/2.-0.5))
  ;denominator
  denominator=2*\sin(!dpi*(float(k)/N-float(tau)/B))
```

```

nz = where(denominator ne 0, count)
wzero = where(denominator eq 0, zerocount)
s(nz) = factor*imag*[firstterm(nz)-secondterm(nz)]/denominator(nz)
if zerocount gt 0 then begin
  s(wzero) = sqrt(b)* \exp(imag*!dpi*tau)
endif
endif else begin
  case order of
  0 : s = complex(1,0) + complexarr(N)
  1: begin
    t = findgen(N)
    s = complex(cos(2*!pi*1*t/N), -\sin(2*!pi*1*t/N))
    end
  endcase
endelse
return, s
end ; end of function

```

The application of the basis functions to calculate the DOST for the positive frequencies is as follows:

```

function dost, timeseries

length = n_elements(timeseries)
partitions = get_partitions(length)
nparts = n_elements(partitions)
sp = ptrarr(nparts/2, /allocate_heap)

; NOTE: positive freqs only
for i = 0L, nparts/2 -1 do begin
  part_len = partitions(i)
  voice = complexarr(part_len)
  for j = 0, part_len - 1 do begin
    basisfunction = basis_definition(length, i, j)/sqrt(float(2^(i-1)))
    dotproduct = total(timeseries * basisfunction)
    voice[j] = dotproduct
  endfor
  *sp(i) = voice
endfor
return, sp
end

```

The partitions can be calculated with the following routine:

```

Function get_partitions, n
n_freqs= long(n/2)
a = partition(n_freqs) ; half the freqs
sum = total(a)
if sum lt n_freqs then a = [a, fltarr(n_freqs-sum)+1]
if n mod 2 eq 0 then begin
  a = [reverse(a), a(1:*)] ; if nyquist, do not double the nyquist
endif else begin
  a = [reverse(a), a(0:*)] ; if no nyquist, double them all
endelse

```

```

a = [1,a] ; add DC to front
return,long(a)
end

```

The recursive function to calculate the partitions is:

```

function partition,n,partsizes=partsizes

if n eq 1 then return,1
if not(keyword_Set(partsizes)) then begin
    if (n mod 2) eq 0 then partsizes = 1
endif
maxpartitionsizes = long(float(n+1)/2)
half = long(float(n+1)/2)
if keyword_set(partsizes) then $
    partsizes=[partsizes,half] $
else partsizes=half ;pass information to recursive call
remainder = n - half
if remainder gt 0L then begin
    a = partition(remainder,partsizes=partsizes)
endif
return,partsizes
end

```

References

- [1] M. Portnoff, Time-frequency representation of digital signals and systems based on short-time Fourier analysis, *IEEE Trans. Acoust. Speech Signal Process.* 28 (1) (1980) 55–69.
- [2] D. Gabor, Theory of communication, *J. Inst. Elect. Eng.* 93 (3) (1946) 429–457.
- [3] P. Bloomfield, *Fourier Analysis of Time Series: An Introduction*, Wiley, New York, 1976.
- [4] L. Cohen, Time-frequency distributions—A review, *Proc. IEEE* 77 (7) (1989) 941–981.
- [5] Y. Zhao, L. Atlas, R. Marks, The use of cone-shaped kernels for generalized time-frequency representations of nonstationary signals, *IEEE Trans. Acoust. Speech Signal Process.* 38 (1990) 1084–1091.
- [6] H. Choi, W. Williams, Improved time-frequency representation of multicomponent signals using exponential kernels, *IEEE Trans. Acoust. Speech Signal Process.* 37 (1989) 862–871.
- [7] F. Hlawatsch, G. Boudreaux-Bartels, Linear and quadratic time frequency signal representations, *IEEE Signal Process. Mag.* (1992) 21–67.
- [8] P. Goupillaud, A. Grossmann, J. Morlet, Cycle-octave and related transforms in seismic analysis, *Geoexploration* 23 (1984) 85–102.
- [9] I. Daubechies, The wavelet transform, time-frequency localization and signal analysis, *IEEE Trans. Inform. Theory* 36 (5) (1990) 961–1005.
- [10] R. Young, *Wavelet Theory and Its Applications*, Kluwer Academic, Dordrecht/Norwell, MA, 1993.
- [11] O. Rioul, M. Vetterli, Wavelets and signal processing, *IEEE Signal Process. Mag.* 8 (1991) 14–38.
- [12] S. Qian, D. Chen, Understanding the nature of signals whose power spectra change with time, *IEEE Signal Process. Mag.* 3 (1999) 51–67.
- [13] J. Lina, Image processing with complex Daubechies wavelet, *J. Math. Imag. Vision* 7 (1996) 211–223.
- [14] N. Kingsbury, Image processing with complex wavelets, *Phil. Trans. R. Soc. London A* 357 (1760) (1997) 2543–2560.
- [15] N. Kingsbury, The dual-tree complex wavelet transform: A new efficient tool for image restoration and enhancement, in: *Proc. EUSIPCO 98*, Rhodes, Greece, 1998, pp. 319–322.
- [16] N. Kingsbury, Shift invariant properties of the dual-tree complex wavelet transform, in: *Proceedings on the International Conference on Acoustics, Speech and Signal Processing*, ICASSP 99, 1999.
- [17] A. Teolis, *Computational Signal Processing with Wavelets*, Birkhäuser, Boston, MA, 1998.
- [18] J. Lina, Complex daubechies wavelets: Filters design and applications, in: *Proc. ISAAC Conf.*, Univ. of Delaware, June 1997, pp. 1–18.
- [19] R. Stockwell, L. Mansinha, R. Lowe, Localization of the complex spectrum: The S-transform, *IEEE Trans. Signal Process.* 44 (4) (1996) 998–1001.
- [20] R. Stockwell, R. Lowe, Airglow imaging of gravity waves—1. Results from a small network of OH nightglow scanning imagers, *J. Geophys. Res. Atmos.* 106 (D15) (2001) 17,185–17,203.
- [21] R. Stockwell, R. Lowe, Airglow imaging of gravity waves—2. Critical layer filtering, *J. Geophys. Res. Atmos.* 106 (D15) (2001) 17,205–17,220.
- [22] H. Liu, R. Roble, A study of a self-generated stratospheric sudden warming and its mesospheric-lower thermospheric impacts using the coupled TIME-GCM/CCM3, *J. Geophys. Res. Atmos.* 107 (D23) (2002) 4695.

- [23] E. Merzlyakov, Y. Portnyagin, C. Jacobi, N. Mitchell, H. Muller, A. Manson, A. Fachrutdinova, W. Singer, P. Hoffmann, On the longitudinal structure of the transient day-to-day variation of the semidiurnal tide in the mid-latitude lower thermosphere—I. Winter season, *Ann. Geophys.* 19 (5) (2001) 545–562.
- [24] D. Pancheva, E. Merzlyakov, N. Mitchell, Y. Portnyagin, A. Manson, C. Jacobi, C. Meek, Y. Luo, R. Clark, W. Hocking, J. McDougall, H. Muller, D. Kurschner, G. Jones, R. Vincent, I. Reid, W. Singer, K. Igarashi, G. Fraser, A. Fahrutdinova, A. Stepanov, L. Poole, S. Malinga, B. Kashcheyev, A. Oleynikov, Global-scale tidal variability during the PSMOS campaign of June–August 1999: Interaction with planetary waves, *J. Atmos. Solar Terrest. Phys.* 64 (17) (2002) 1865–1896.
- [25] Y. Portnyagin, E. Merzlyakov, C. Jacobi, N. Mitchell, H. Muller, A. Manson, W. Singer, P. Hoffmann, A. Fachrutdinova, Some results of S-transform analysis of the transient planetary-scale wind oscillations in the lower thermosphere, *Earth Planets Space* 51 (7–8) (1999) 711–717.
- [26] D. Fritts, D. Riggan, B. Balsley, R. Stockwell, Recent results with an MF radar at McMurdo, Antarctica: Characteristics and variability of motions near 12-hour period in the mesosphere, *Geophys. Res. Lett.* 25 (3) (1998) 297–300.
- [27] L. Mansinha, R. Stockwell, R. Lowe, M. Eramian, R. Schincariol, Local S-spectrum analysis of 1-D and 2-D data, *Phys. Earth Planet. Inter.* 103 (3–4) (1997) 329–336.
- [28] J. Gao, W. Chen, Y. Li, F. Tian, Generalized S-transform and seismic response analysis of thin interbeds, *Chin. J. Geophys.* 46 (4) (2003) 526–532 [in Chinese].
- [29] S. Theophanis, J. Queen, Color display of the localized spectrum, *Geophysics* 65 (4) (2000) 1330–1340.
- [30] J. Osler, D. Chapman, Seismo-acoustic determination of the shear-wave speed of surficial clay and silt sediments on the Scotian shelf, *Canad. Acoust.* 24 (3) (1996) 11–22.
- [31] P. Chu, A high resolution global sea surface temperature analysis and climatology, *Trop. Oceans Global Atmos. Notes* 17 (1994) 5–7.
- [32] P. Chu, The S-transform for obtaining localized spectra, *J. Marine Technol. Soc.* 29 (4) (1996) 28–38.
- [33] P. Dash, B. Panigrahi, G. Panda, Power quality analysis using S-transform, *IEEE Trans. Power Deliv.* 18 (2) (2003).
- [34] Chilukuri, Multiresolution S-transform based fuzzy recognition system for power quality event, *IEEE Trans. Power Deliv.* 19 (1) (2003) 323–330.
- [35] P. McFadden, J. Cook, L. Forster, Decomposition of gear vibration signals by the generalised S-transform, *Mech. Syst. Signal Process.* 13 (5) (1999) 691–707.
- [36] P. McFadden, Detection of gear faults by decomposition of matched differences of vibration signals, *Mech. Syst. Signal Process.* 14 (5) (2000) 805–817.
- [37] C. Pinnegar, L. Mansinha, The bi-Gaussian S-transform, *SIAM J. Sci. Comput.* 24 (5) (2003) 1678–1692.
- [38] C. Pinnegar, L. Mansinha, The S-transform with windows of arbitrary and varying shape, *Geophysics* 68 (1) (2003) 381–385.
- [39] C. Pinnegar, L. Mansinha, Time-frequency localization with the Hartley S-transform, *Signal Process.* 84 (12) (2004) 2437–2442.
- [40] C. Pinnegar, L. Mansinha, Time-local Fourier analysis with a scalable, phase-modulated analyzing function: The S-transform with a complex window, *Signal Process.* 84 (7) (2004) 1167–1176.
- [41] S. Andino, R. Menendez, C. Lantz, O. Blank, C. Michel, T. Landis, Non-stationary distributed source approximation: An alternative to improve localization procedures, *Human Brain Mapp.* 14 (2) (2001) 81–95.
- [42] M. Varanini, G. Paolis, M. Emdin, A. Macerata, S. Pola, M. Cipriani, C. Marchesi, Spectral analysis of cardiovascular time series by the S-transform, *Comput. Cardiol.* 24 (1997) 383–386.
- [43] B. Goodyear, H. Zhu, R. Brown, J. Mitchell, Removal of phase artifacts from fMRI data using a Stockwell transform filter improves brain activity detection, *Magn. Reson. Med.* 51 (1) (2003) 16–21.
- [44] S. Assous, A. Humeau, M. Tartas, P. Abraham, J. L'Huillier, Physiological effects of indomethacin and celecoxib: An S-transform laser Doppler flowmetry signal analysis, *Phys. Med. Biol.* 50 (2005) 1951–1959.
- [45] X.-P. Zhang, M. Desai, Y. Peng, Orthogonal complex filter banks and wavelets: Some properties and design, *IEEE Trans. Signal Process.* 47 (4) (1999) 1039–1048.
- [46] I. Selesnick, L. Sendur, Iterated oversampled filter banks and wavelet frames, in: *Proc. SPIE Wavelet Appl. Signal Image Process.*, vol. VIII, 2000, p. 4119.
- [47] D. Nelson, Cross-spectral methods for processing speech, *J. Acoust. Soc. Amer.* 110 (2001) 2575–2592.
- [48] T. Gardner, M. Magnasco, Instantaneous frequency decomposition: An application to spectrally sparse sounds with fast frequency modulations, *J. Acoust. Soc. Amer.* 117 (5) (2005) 2896–2903.
- [49] R. Bracewell, *The Fourier Transform and Its Applications*, McGraw-Hill, New York, 1978.
- [50] Y. Hayashi, Space-time spectral analysis of rotary vector series, *J. Atmos. Sci.* 36 (5) (1979) 757–766.
- [51] R. Stockwell, W.G. Large, R. Milliff, Resonant inertial oscillations in moored buoy ocean surface winds, *Tellus A* 56 (5) (2004) 546–547.

Dr. R.G. Stockwell is a research scientist at Colorado Research Associates, Boulder, CO, USA, with a strong background in signal processing applied to atmospheric science. He has worked on the analysis of MF radar wind data to characterize semidiurnal oscillations in mesospheric winds, and inertial oscillations in ocean surface winds to analyze their transfer of energy and momentum into the ocean at resonant frequencies. Dr. Stockwell has published works on the analysis of airglow data and on modeling of gravity wave propagation characteristics.